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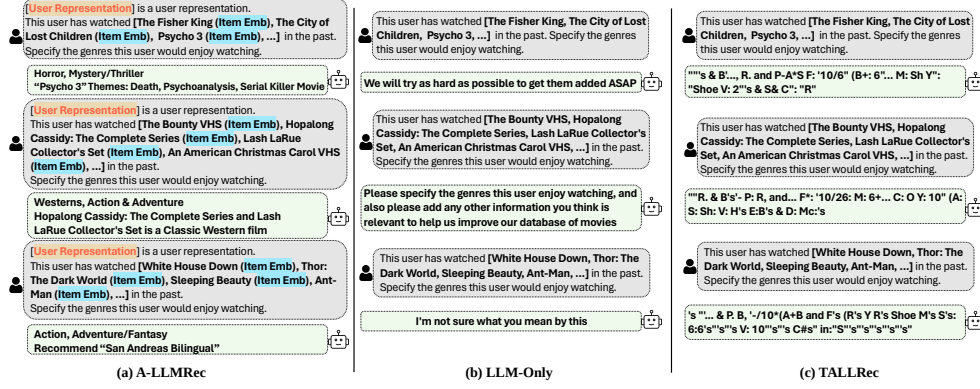


Figure 5: A-LLMRec, LLM-Only, and TALLRec on the favorite genre prediction task (Movies and TV dataset used).

A BASELINES

(1) Collaborative filtering recommender systems

- **NCF** [15] combines neural networks (MLP) to capture the collaborative information. Note that NCF is a two-tower model comprised of separate components for the user and item embedding matrix.
- **NextItNet** [50] proposes a temporal convolutional network that utilizes 1D-dilated convolutional layers and residual connections to capture the long-term dependencies inherent in interaction sequence.
- **GRU4Rec** [17] adopts RNNs to model user behavior sequences for session-based recommendations.
- **SASRec** [20] is our main baseline, a state-of-the-art collaborative filtering recommender system (CF-RecSys) that adopts a self-attention encoding method to model user preferences from user behavior sequences.

(2) Modality-aware recommender systems

- **MoRec** [51] employs a pre-trained SBERT to utilize the text information of items to generate the initial embeddings for items that will be used in collaborative filtering models. We utilize SASRec as the backbone model of MoRec.
- **CTRL** [25] employs a two-stage learning process: the first stage involves contrastive learning on textual information of items to initialize the backbone model, and the second stage, fine-tunes the model on recommendation tasks. We use SASRec as the backbone model of CTRL.
- **RECFORMER** [24] models user preferences and item features using the Transformer architecture, transforming sequential recommendation into a task of predicting the next item as if predicting the next sentence, by converting item attributes into a sentence format.

(3) LLM-based recommender systems

- **LLM-Only** utilizes an open-source LLM model OPT [52] with prompts related to recommendation tasks as shown in Figure 6. In our experiments, we adopt the 6.7B size version of OPT for all LLM-based recommendations.
- **TALLRec** [2] is our main baseline, which learns the recommendation task based on prompts consisting solely of text and fine-tunes the LLMs using the LoRA. Their approach involves providing user interaction history and one target item and determining whether a user will prefer this target item. This simpler task necessitates only a brief prompt for the LLMs.

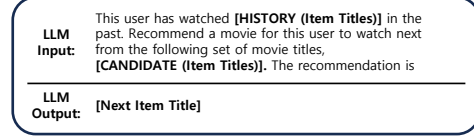


Figure 6: An example prompt designed for the Amazon Movies dataset used by LLM-based models, i.e., TALLRec and LLM-Only models.

Table 12: Source code links of the baseline methods.

Methods	Source code
SASRec	https://github.com/pmixer/SASRec.pytorch
NextItNet	https://github.com/syiswell/NextItNet-Pytorch
GRU4Rec	https://github.com/hungphanh/GRU4REC-pytorch
RECFORMER	https://github.com/AaronHeee/RecFormer
TALLRec	https://github.com/SAI990323/TALLRec
A-LLMRec	https://github.com/ghdtjr/A-LLMRec

In contrast, our recommendation task requires a more extensive prompt. Even though this adjustment results in a smaller batch size, the same as A-LLMRec, for training TALLRec. We use the prompt shown in Figure 6.

- **MLP-LLM** is an additionally designed LLM-based recommendation model for analysis. Compared with A-LLMRec, this model directly connects the user and item embeddings from frozen CF-RecSys and LLM using only MLP layers, instead of the auto-encoders in A-LLMRec that involve various techniques to align the collaborative knowledge of CF-RecSys with the LLM. Note that we use the prompt shown in Figure 3.

B LANGUAGE GENERATION TASK

In Figure 5, we present additional favorite genre prediction task results for experiment in shown in Section 5.4.4. As mentioned in Section 5.4.4, TALLRec could not generate valid natural language outputs due to the fine-tuning via instruction tuning process, which makes the LLM of TALLRec being able to answer only with some particular prompts used in instruction tuning process. The additional results indicate that A-LLMRec can generate the favorite genres for the users based on the understanding of the aligned user representation and item embeddings while LLM-only fails to do so.

C REPRODUCIBILITY

For implementing the baseline, we followed the official codes published by authors as detailed in Table 12. Refer to our source code and instructions to run code for reproducing the results reported in the experiments.